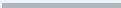




GOOGLE SUMMER OF CODE 2026

Improve Getting Started Documentation

Project Proposal by SammanSarkar



Applicant: SammanSarkar

Project Period: May - August 2026

Submission Date: March 2026



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1. Project Overview

1.1 Background

dora-rs is a high-performance dataflow framework for building robotic applications. As the project evolves, the learning curve for new users has become a bottleneck for community growth. This project aims to create systematic getting-started documentation, video tutorials, and example code to lower the entry barrier and help developers quickly grasp the core concepts and usage of dora-rs and adora.

1.2 Objectives

Core Objective

Create a comprehensive learning system for dora-rs and adora projects, including documentation tutorials, video tutorials, and runnable example code, enabling new users to complete the journey from installation to running their first complete example within 2-3 hours.

Specific objectives include:

- Write clear, step-by-step getting-started documentation
- Produce accompanying video tutorials (Bilibili and YouTube)
- Provide complete, runnable example code
- Demonstrate practical use cases combining dora-rs with AI tools

2. Applicant Strengths

I have made substantial contributions to the dora-rs ecosystem prior to this application, with the following advantages:

Table 1 Contribution History

Project	Contribution	Link
dora-examples	Example code contributions	https://gitcode.com/dora-org/dora-examples
dora-rs-org	Project website contributions	https://github.com/dora-rs-org/dora-rs-org
Video Tutorial	Beginner-friendly tutorial	https://www.bilibili.com/video/BV1gzzHBCEFn

Video Tutorial Highlights:

- Based on the latest Windows 11 environment
- Demonstrates using AI tools (Claude Code) for programming and debugging
- Starts from scratch, suitable for complete beginners

3. Technology Stack

This project involves the following technologies and tools:

Table 2 Technology Stack

Technology/Tool	Purpose
dora-rs	Core robotic dataflow framework
adora	High-level wrapper for dora-rs
rerun	Visualization and simulation tool
Python	Primary development language
Rust	Performance-critical modules
PowerShell	Windows environment scripting
Claude Code	AI-assisted programming tool

4. Timeline

4.1 Community Bonding Period (May 1 - May 24)

Time	Activities
May 1-7	Establish contact with mentors, clarify project details and expectations
May 8-14	Deep dive into dora-rs and adora documentation, understand latest developments
May 15-21	Start handling small contributions, familiarize with code submission workflow
May 22-24	Develop detailed Phase 1 work plan

4.2 Coding Phase 1 (May 25 - July 10)

Coding officially begins: May 25, 2026

4. Timeline

Week	Date Range	Main Tasks
Week 1	May 25-31	dora-rs and adora introduction docs, installation guide and Hello World example
Week 2	June 1-7	rerun simulation tool introduction, create first simple scene
Week 3	June 8-14	Claude Code usage tutorial, write example code driving rerun scene
Week 4	June 15-21	Retrieve camera data from rerun, implement visual feedback
Week 5	June 22-28	Consolidate previous content, refine documentation and videos
Week 6	June 29 - July 5	Midterm evaluation preparation, submit work products

Midterm evaluation deadline: July 10, 2026 18:00 UTC

4.3 Coding Phase 2 (July 11 - August 24)

Week	Date Range	Main Tasks
Week 7	July 11-17	Integrate open-source multimodal models, analyze scene visual information
Week 8	July 18-24	Acquire more sensor data, expand scene interaction capabilities
Week 9	July 25-31	Plan action paths using LLM large models
Week 10	August 1-7	Integrate agentsdk for automated task planning
Week 11	August 8-14	adora + octos advanced integration (ultimate intelligent experience project)
Week 12	August 15-21	Final consolidation, complete all documentation and videos
Final Week	August 22-24	Submit final work product and mentor evaluation

Final submission deadline: August 24, 2026 18:00 UTC

5. Detailed Work Plan

5.1 Pre-Midterm Goals

1

dora-rs and adora Introduction

Write complete installation guide covering Windows, Linux, and macOS platforms.
Create Hello World example demonstrating basic dataflow node communication.

2

rerun Simulation Introduction

Introduce basic concepts and usage of rerun visualization tool. Create a simple simulation scene (e.g., vehicle or robotic arm) as foundation for subsequent tutorials.

3 Claude Code Usage Tutorial

Write tutorial on using Claude Code for dora-rs development. Create example code demonstrating AI-driven vehicle/arm in rerun scene.

4 Camera Data Acquisition

Implement camera data retrieval from rerun scene. Demonstrate integrating visual data into dora-rs dataflow.

5.2 Pre-Final Goals

5 Multimodal Model Integration

Integrate open-source multimodal models (e.g., LLaVA, Qwen-VL) to analyze scene visual information. Expand sensor data acquisition capabilities (depth, pose, etc.).

6 LLM Action Planning

Leverage large language models for task understanding and action planning. Implement conversion from natural language instructions to robot actions.

7 agentsdk Automated Task Planning

Integrate agentsdk framework for automated task decomposition and execution. Create automated planning examples for multi-step tasks.

8 adora + octos Advanced Integration

Explore deep integration of adora with octos for more intelligent robot control. Create ultimate intelligent experience demonstration project.

6. Deliverables

All deliverables will be published in the following formats:

Table 3 Deliverables List

Type	Platform	Content
Documentation	GitHub / GitCode	Complete tutorial documentation in Markdown format
Example Code	GitHub / GitCode	Runnable code for each milestone
Video Tutorial	Bilibili	Chinese video tutorials accompanying documentation
Video Tutorial	YouTube	English video tutorials for international users

7. Risk Assessment

Table 4 Risk Assessment

Risk	Likelihood	Mitigation
Technical dependency updates causing compatibility issues	Medium	Regularly sync upstream updates, maintain communication with mentors
Multimodal model integration complexity exceeds expectations	Medium	Prepare simplified alternatives, prioritize core functionality
Insufficient time for video production	Low	Prioritize documentation quality, videos can be released in phases
Personal time conflicts	Low	Plan ahead, sync progress with mentors weekly

8. Reference Links

- dora-examples project: <https://gitcode.com/dora-org/dora-examples>
- dora-rs-org project: <https://github.com/dora-rs-org/dora-rs-org>
- Getting Started Video Tutorial: <https://www.bilibili.com/video/BV1gzzHBCEFn>
- GSoC 2026 Official Timeline: <https://summerofcode.withgoogle.com/>

Commitment

I commit to completing this project according to the above plan, maintaining regular communication with my mentor, and ensuring the quality of deliverables. Any changes to the plan will be discussed with my mentor in advance and approved.